

2. Carbohydrates

Learning objectives

At the end of this session the students will be able to:-

- Explain what is meant by the terms monosaccharide, disaccharide, oligosaccharide, and polysaccharide.
- Describe the formation of glycosides and the structures of the important disaccharides and polysaccharides.
- Explain the roles of carbohydrates.

Definitions

- **Glycobiology** is the study of the roles of sugars in health and disease.
- The **glycome** is the entire complement of sugars of an organism, whether free or present in more complex molecules.
- **Glycomics**, an analogous term to genomics and proteomics, is the comprehensive study of glycomes, including genetic, physiological, pathological, and other aspects.

Sources of Carbohydrates



Carbohydrates

- are the most abundant organic molecules in nature.

Biomedical Importance(functions)

1. source and storage of energy, glucose and glycogen respectively.
2. Structural components. E.g. skin, bone, cell mm
3. Involved in cell-cell interaction
4. Their derivatives are drugs. E.g. erythromycin
5. Survival of Antarctic fish in icy environment is due to presence of anti-freeze glycoprotein's in their blood.
6. Ascorbic acid, a derivative of carbohydrate is a water-soluble vitamin..

Carbohydrates

Metabolic / Nutritional

- principal part of the energy
- Carbohydrate intake can take place in different forms like sugar, starch, fibers etc.
- Fiber does wonders in keeping your bowel function going smooth.
- Carbohydrates add on to the taste and appearance of food item, thus making the dish tempting and mouth-watering
- They are sometimes used as flavours and sweeteners.
- Carbohydrates aid in regulating blood glucose and also do good to our body by breaking down fatty acids, thus preventing ketosis.

Chemical Nature of Carbohydrates

- They are polyhydroxy alcohols with a functional aldehyde or keto group.
 - Most sugars have formula $C_n(H_2O)_n$, “hydrate of carbon.”

Classification of Carbohydrates

- based on number of carbon chains present; they are:
 1. **Monosaccharides** - simple sugars with multiple OH groups. Based on number of carbons (3, 4, 5, 6), a monosaccharide is a *triose*, *tetrose*, *pentose* or *hexose*.
 2. **Disaccharides** - 2 monosaccharides covalently linked.
 3. **Oligosaccharides** - a few monosaccharides covalently linked. (3-10 monomer)
 4. **Polysaccharides** - polymers consisting of chains of monosaccharide or disaccharide units. (>10 monomer)

Types of Carbohydrates

Simple Carbohydrates

- monosaccharides
- disaccharides

Complex Carbohydrates

- oligosaccharides
- polysaccharides
 - glycogen
 - starches
 - Fibers

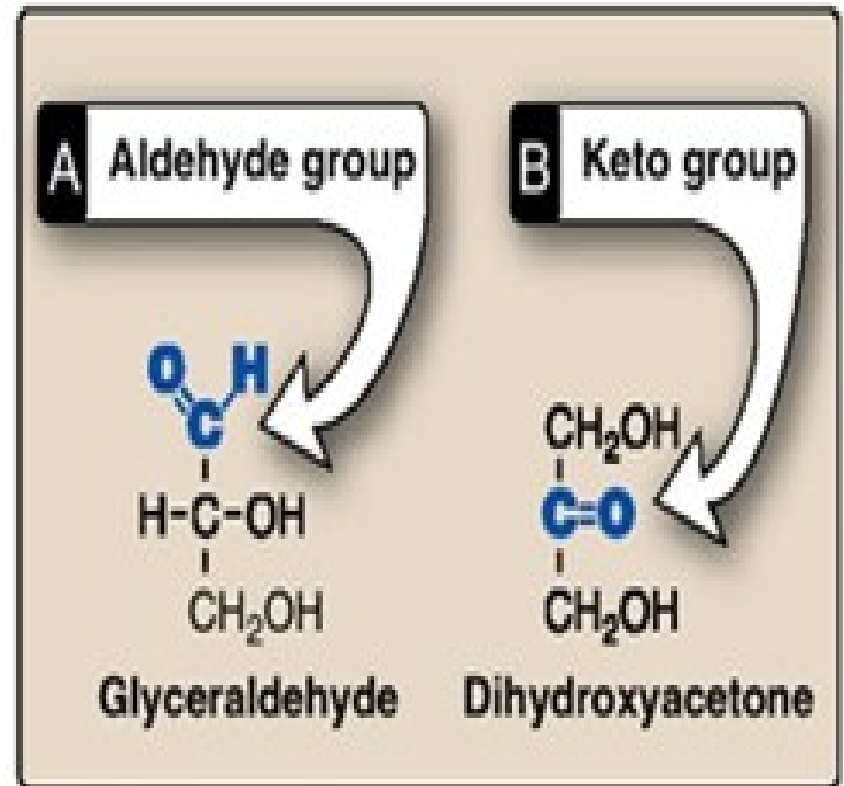
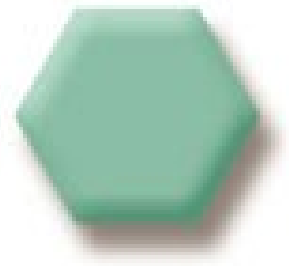


Fig. examples of an aldose (A) and a ketose(B) sugar.

1. Monosaccharides: Single Sugars



Glucose

Glucose

- carbohydrate form used by the body, referred to as “blood sugar”
- basic sub-unit of other larger carbohydrate molecules
- found in fruits, vegetables, honey
- The oxidation of glucose by **glucose oxidase** (a highly specific test for glucose) is used by clinical and other laboratories to measure the amount of **glucose in urine** using a dipstick.

Monosaccharides: Single Sugars

Fructose

- sweetest of the sugars
- occurs naturally in fruits & honey, “fruit sugar”
- combines with glucose to form sucrose



Fructose

Galactose

- combines with glucose to form lactose, “milk sugar”



Galactose

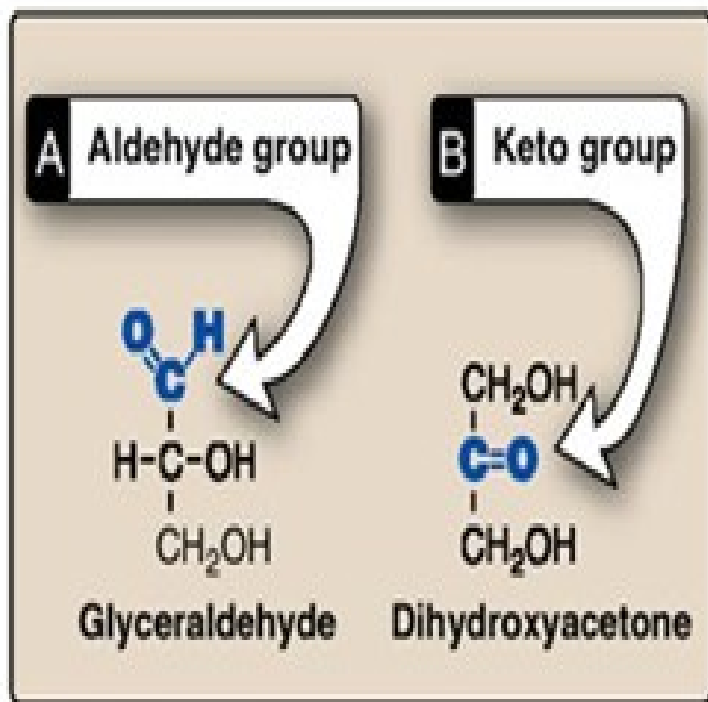
1. Monosaccharides

- They can not be hydrolyzed to small compounds.
- Their general formula is $C_n(H_2O)_n$.
- They are also called as *simple sugars*.

Nomenclature

- have *common (trivial) names* and *systematic names*.
- Systematic name indicates both the number of carbon atoms present and aldehyde or ketone group.
- ✓ E.g. **glyceraldehyde** is a simple sugars containing three carbon atoms and a aldehyde group.
- Simple sugars containing **three** carbon atoms are referred as **trioses**.
- In addition, sugars containing aldehyde group or keto group are called as **aldoses** or **ketoses**, respectively.

- Thus, the systematic name for glyceraldehyde is *aldotriose*.
- Like wise a simple sugar with three carbon atoms and a keto group is called as *ketotriose*.



<u>Generic names</u>	<u>Examples</u>
3 Carbons: trioses	Glyceraldehyde
4 Carbons: tetroses	Erythrose
5 Carbons: pentoses	Ribose
6 Carbons: hexoses	Glucose
7 Carbons: heptoses	Sedoheptulose
9 Carbons: nonoses	Neuraminic acid

Fig. examples of an
aldose (A) and a
ketose(B) sugar.

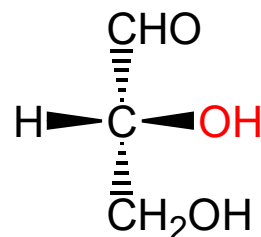
Properties of monosaccharides

1. Optical Isomerism

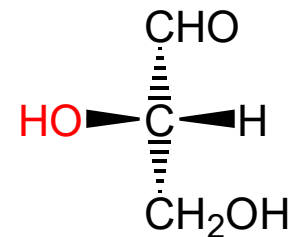
- All the monosaccharides **except dihydroxyacetone** contain at least one asymmetric carbon atom and hence they exhibit optical isomerism.
- D and L-glyceraldehyde are used as parent compounds to designate all other sugars (compounds) as D or L forms.

D & L designations are based on the configuration about the **single asymmetric C** in glyceraldehyde.

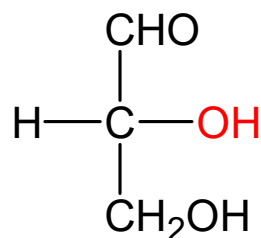
The lower representations are **Fischer Projections**.



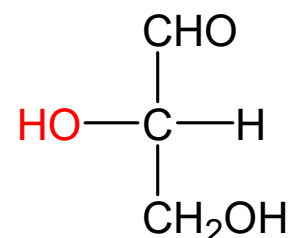
D-glyceraldehyde



L-glyceraldehyde

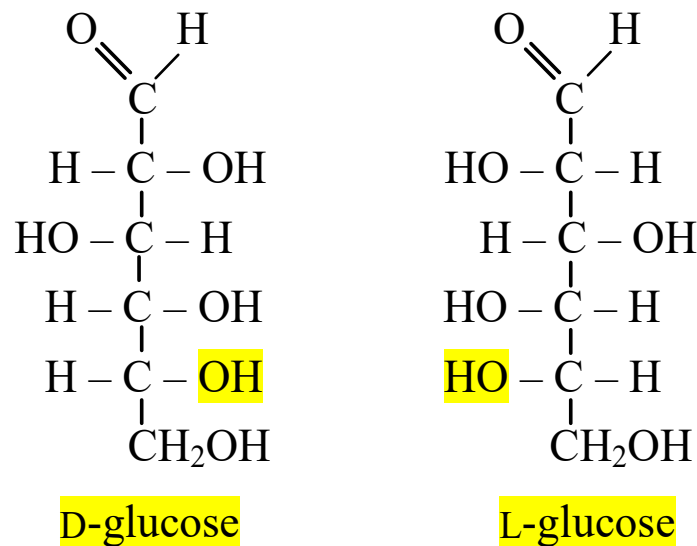


D-glyceraldehyde



L-glyceraldehyde

For sugars with more than one chiral center, **D** or **L** refers to the asymmetric **C** farthest from the aldehyde or keto group.



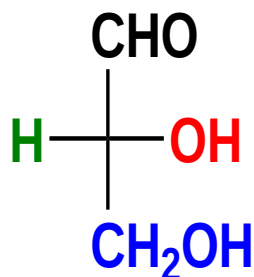
If the hydroxyl group of the highest numbered chiral carbon is pointing to the right, the sugar is designated as **D** (*Dextro*: Latin for *on the right side*).

If the hydroxyl group is pointing to the left, the sugar is designated as **L** (*Levo*: Latin for *on the left side*).

Most naturally occurring carbohydrates are of the D-configuration

2.Optical Activity

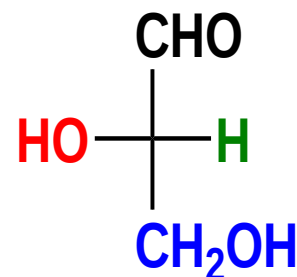
- Monosaccharides **except dihydroxy acetone** exhibit optical activity because of the presence of at least 1 asymmetric carbon atom.
- If a sugar rotates plane polarized light to right then it is called as *dextrorotatory* and if a sugar rotates the plane polarized light to the left then it is called as *levorotatory*.
- Usually '+' sign indicates dextrorotation and '-' sign indicates levorotation of a sugar.



D-glyceraldehyde

R-(+)-glyceraldehyde

(+)-rotation = dextrorotatory



L-glyceraldehyde

S-(-)-glyceraldehyde

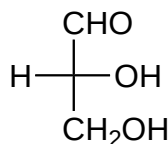
(-)-rotation = levorotatory

The Aldotetroses

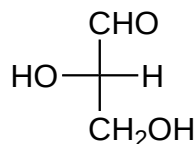
Glyceraldehyde is the simplest carbohydrate (C_3 , aldotriose, 2,3-dihydroxypropanal).

The next carbohydrate are aldotetroses (C_4 , 2,3,4-trihydroxybutanal).

2^n

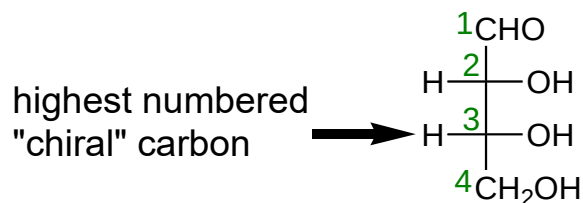


D-glyceraldehyde

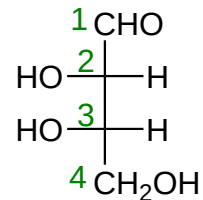


L-glyceraldehyde

aldotetroses

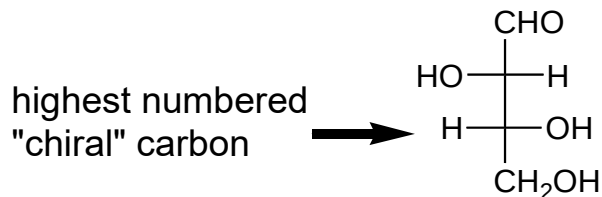


D-erythrose

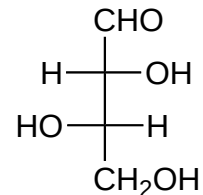


L-erythrose

$\leftarrow$ highest numbered
"chiral" carbon



D-threose

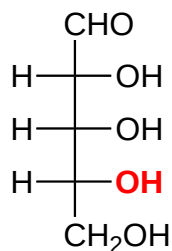


L-threose

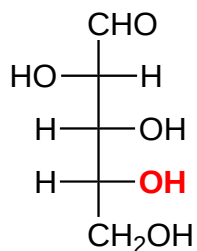
$\leftarrow$ highest numbered
"chiral" carbon

Aldopentoses and Aldohexoses

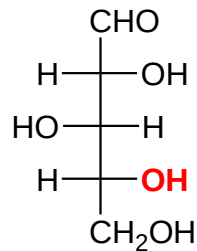
Aldopentoses: C₅, three chiral carbons, eight stereoisomers



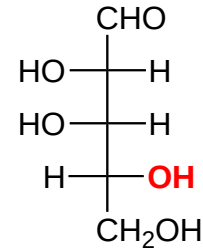
D-ribose



D-arabinose

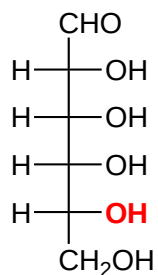


D-xylose

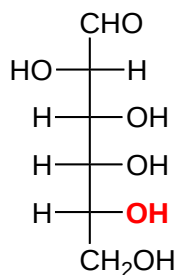


D-lyxose

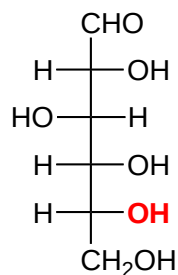
Aldohexoses: C₆, four chiral carbons, sixteen stereoisomers



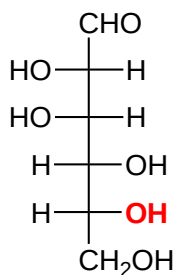
D-allose



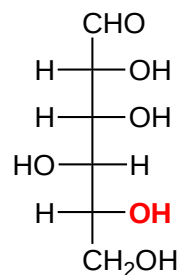
D-altrose



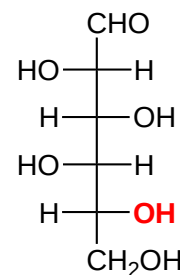
D-glucose



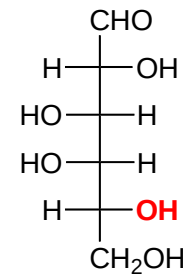
D-mannose



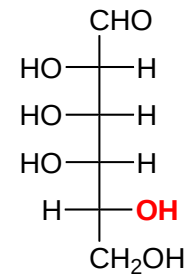
D-gulose



D-idose



D-galactose



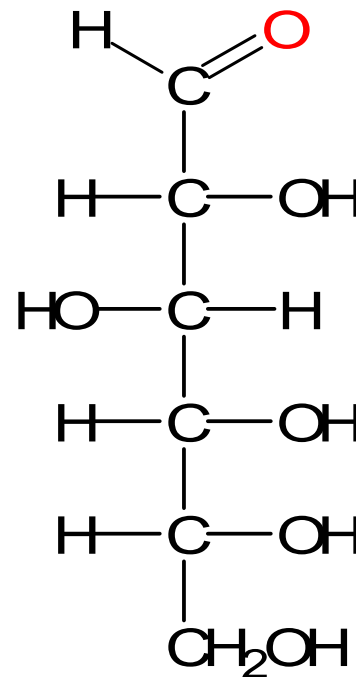
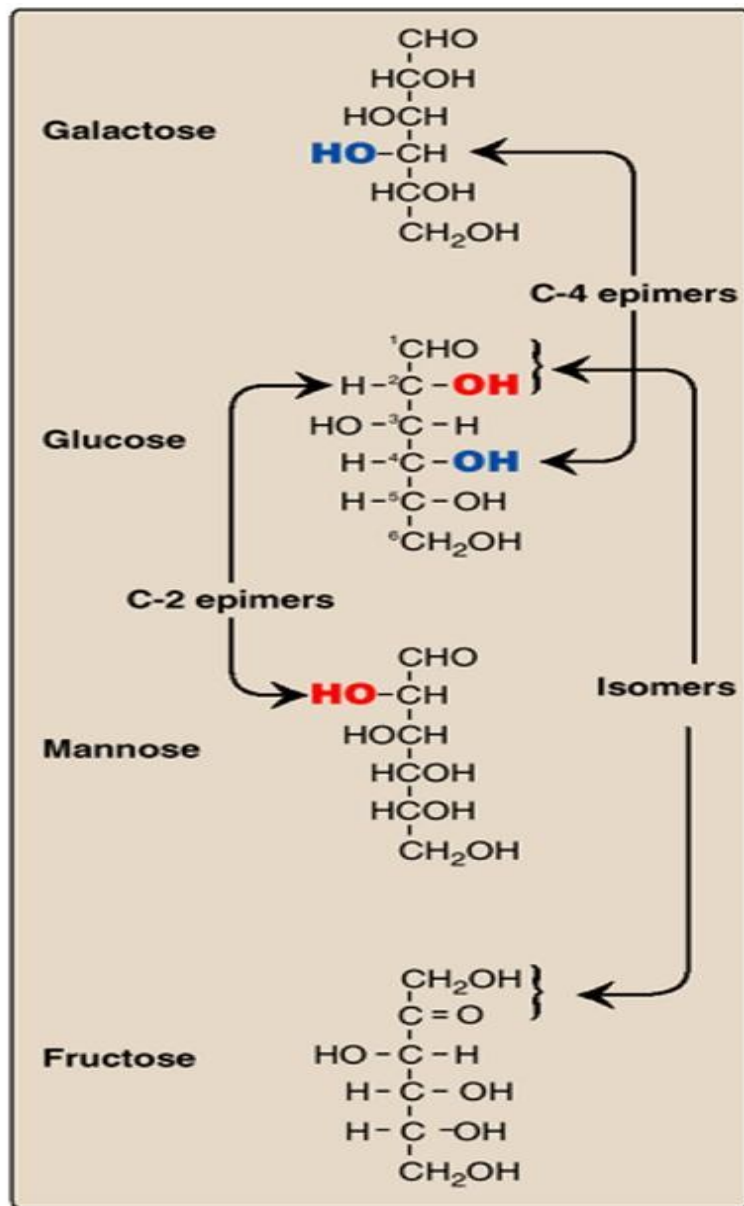
D-talose

3. Isomers and epimers

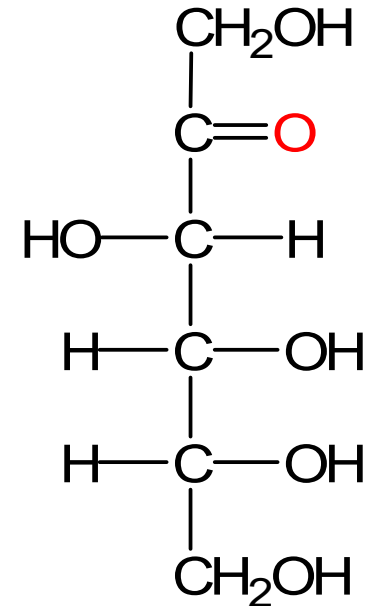
- ✓ Compounds that have the same chemical formula but have different structures are called isomers.
- ✓ E.g. **fructose, glucose, mannose, and galactose** are all isomers of each other, having the same chemical formula, **$C_6H_{12}O_6$** .
- ✓ Carbohydrate isomers that differ in configuration around only one specific carbon atom (with the exception of the carbonyl carbon) are defined as *epimers* of each other.

4. Functional Isomers

- same molecular formulae but different functional groups.
- For example, glucose and fructose have same molecular formulae $C_6H_{12}O_6$, but glucose contains aldehyde as functional group and fructose contains keto group.



D-glucose



D-fructose

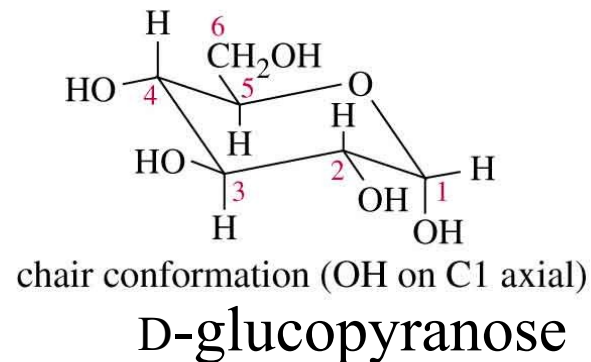
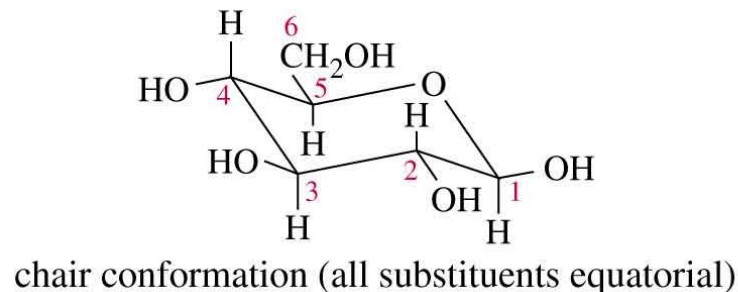
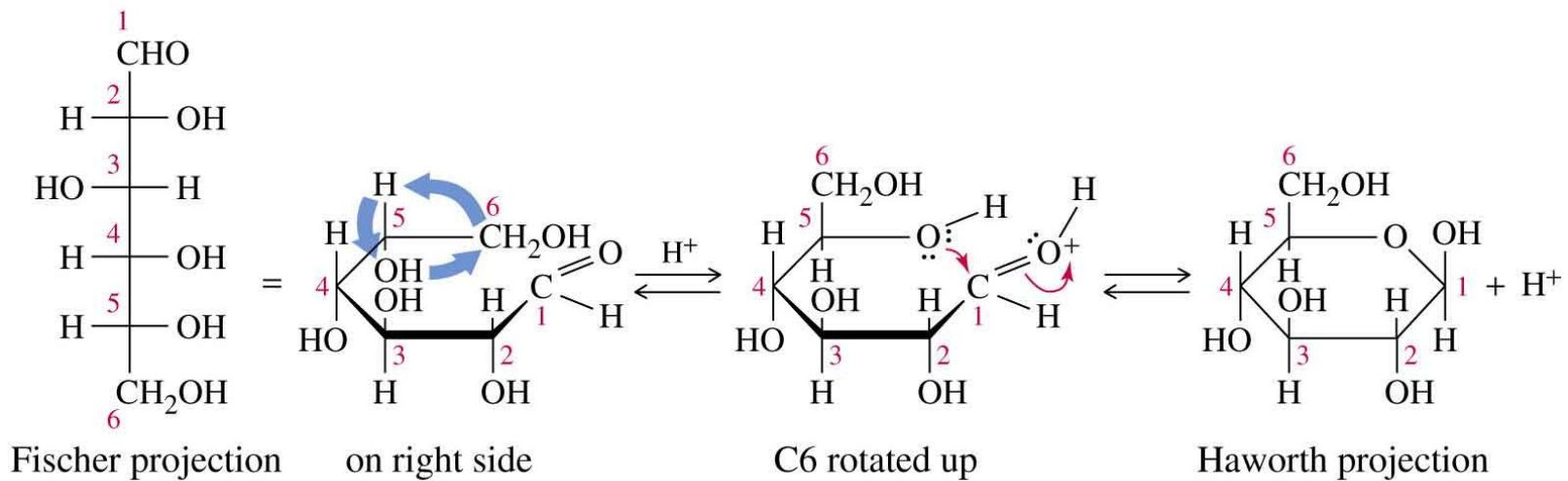
Aldoses (e.g., glucose) have an **aldehyde** group at one end.

Ketoses (e.g., fructose) have a **keto** group, usually at C2.

Figure 7.4 C-2 and C-4 epimers and an isomer of glucose.

Cyclic Structure for Glucose

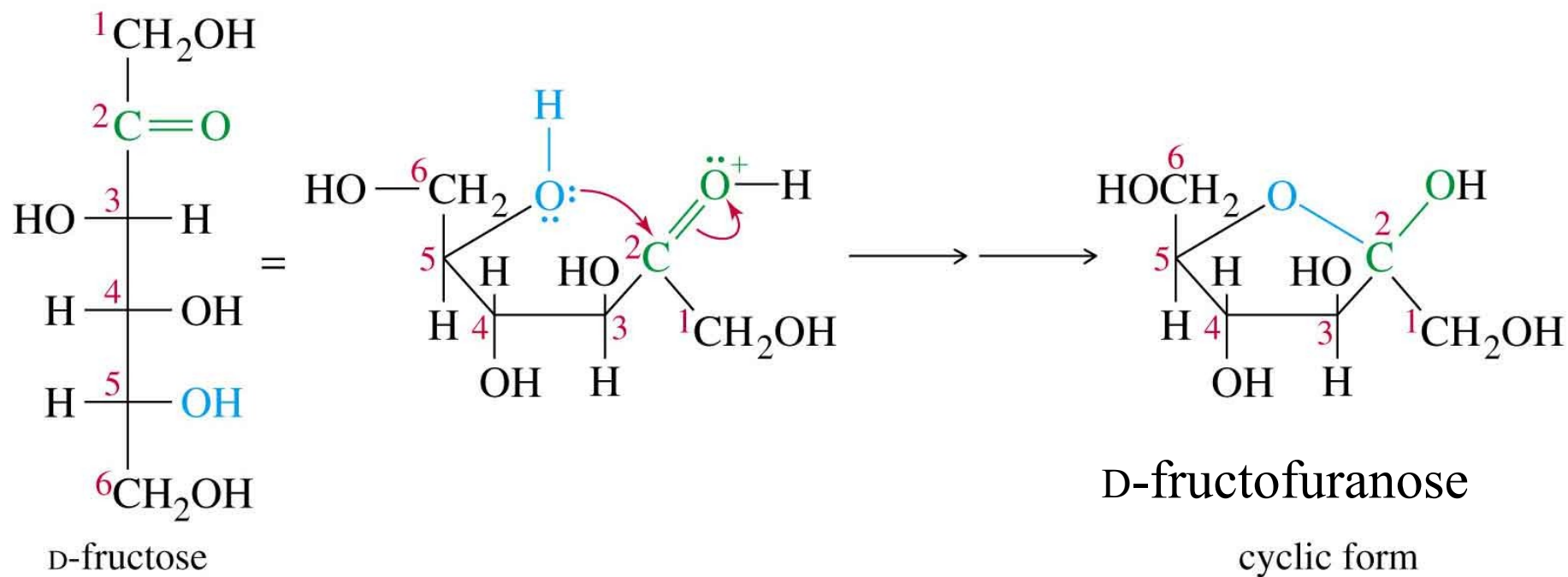
Glucose cyclic hemiacetal formed by reaction of -CHO with -OH on C5.



$\Rightarrow$

Cyclic Structure for Fructose

Cyclic hemiketal formed by reaction of C=O at C2 with -OH at C5.

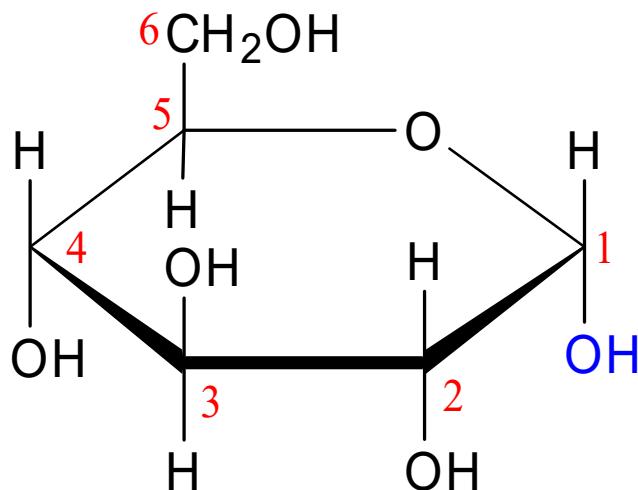


5. Anomers

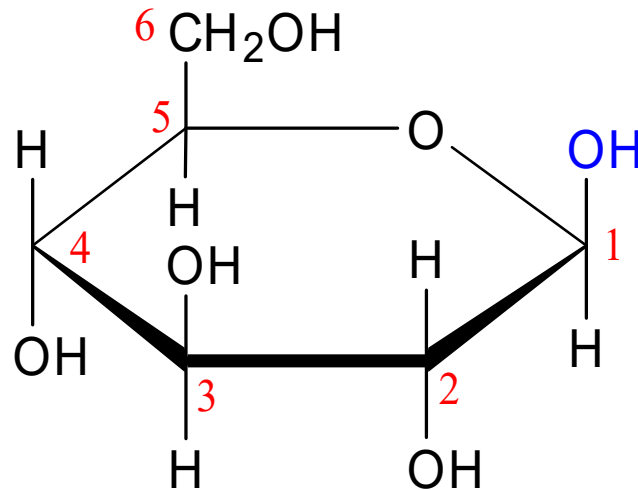
- Cyclization of glucose produces a new **asymmetric center** at C_1 .
- The 2 stereoisomers are called **anomers**, α & β .

Haworth projections represent the cyclic sugars as having essentially planar rings, with the OH at the anomeric C_1 :

- ♦ α (OH **below** the ring)
- ♦ β (OH **above** the ring).



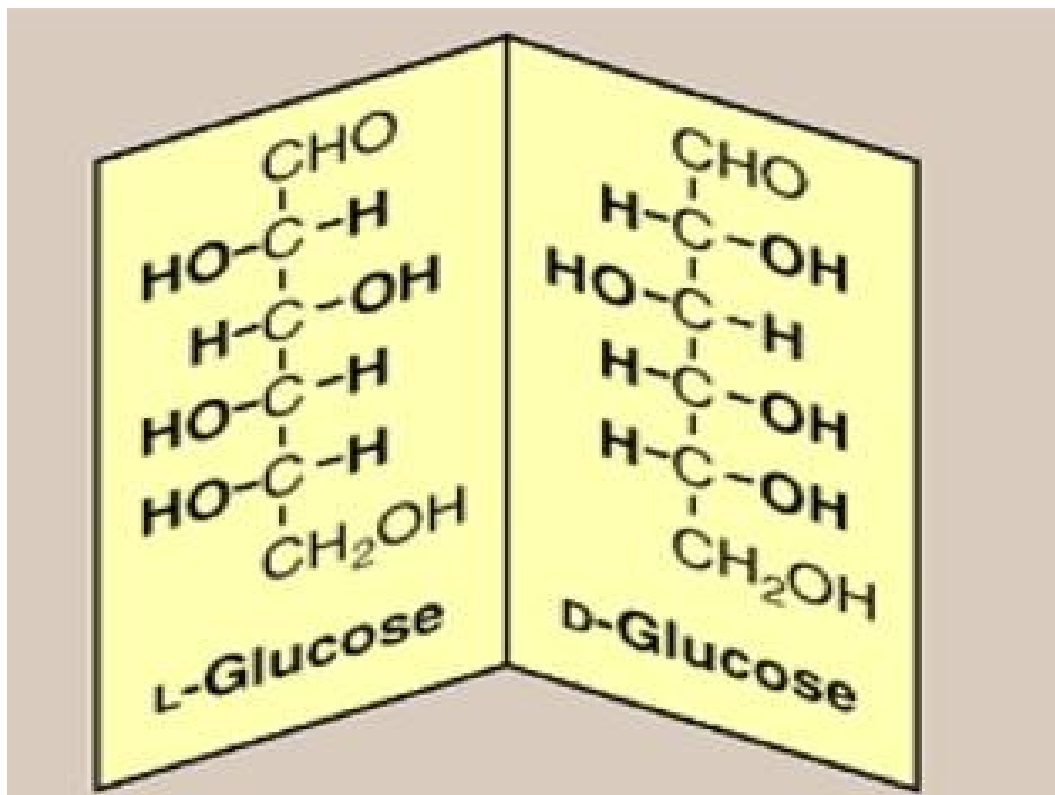
α -D-glucose



β -D-glucose

6. Enantiomers

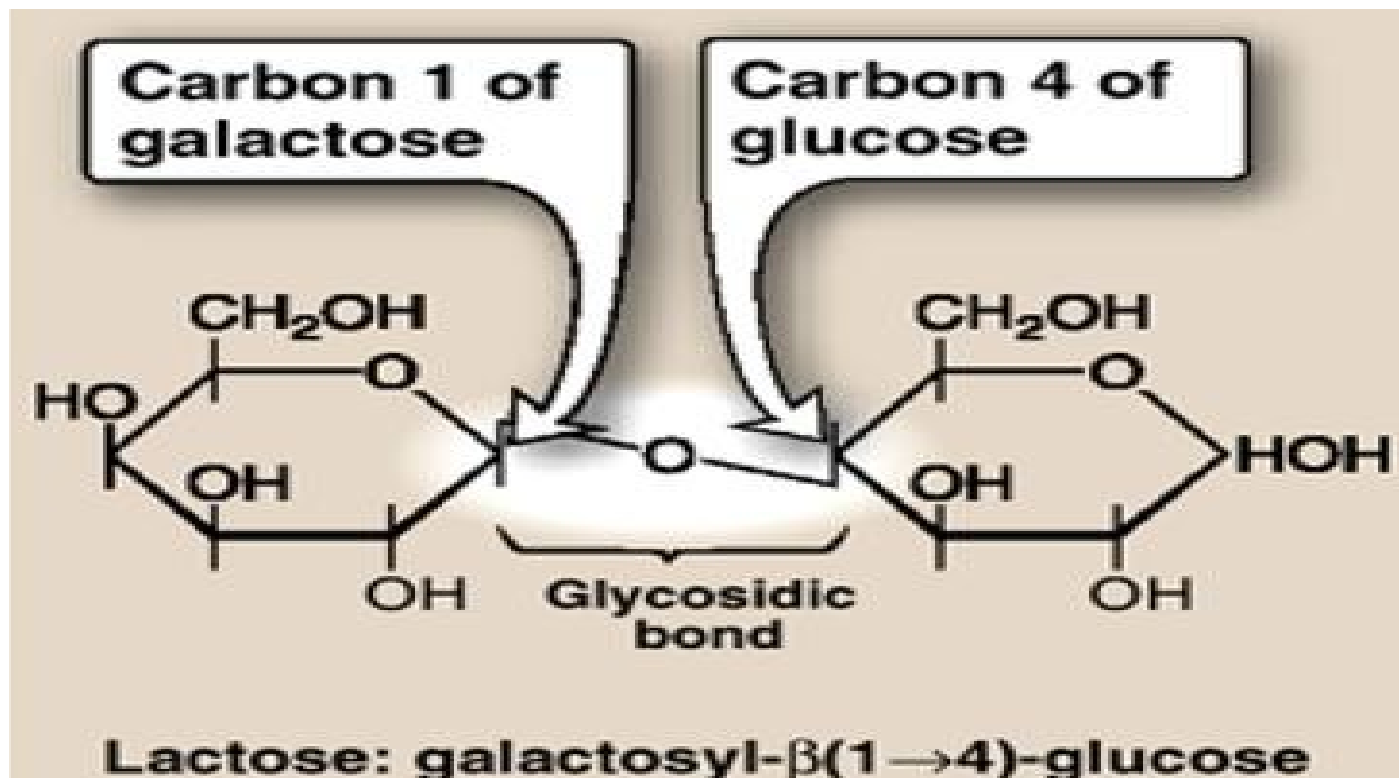
- A special type of isomerism is found in the pairs of structures that are *mirror images of each other*.
- These mirror images are called **enantiomers**.



Enantiomers
(mirror images)
of glucose.

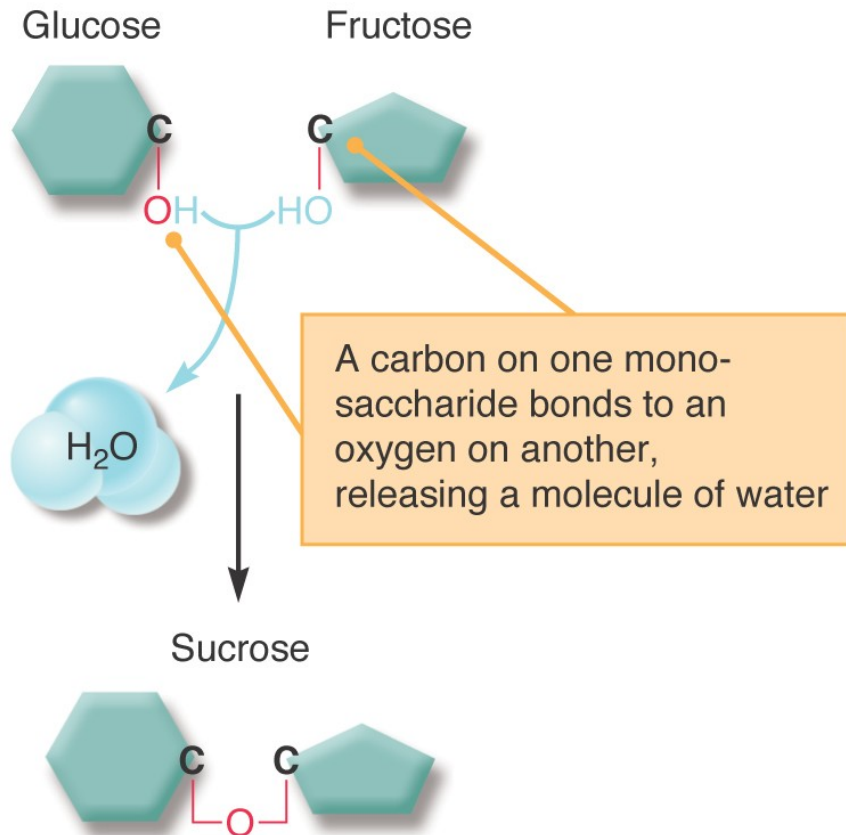
Joining of monosaccharides

- Monosaccharides can be joined to form disaccharides, oligosaccharides, and polysaccharides.
- The bonds that link sugars are called *glycosidic bonds*. These are formed by enzymes known as *glycosyltransferases*.

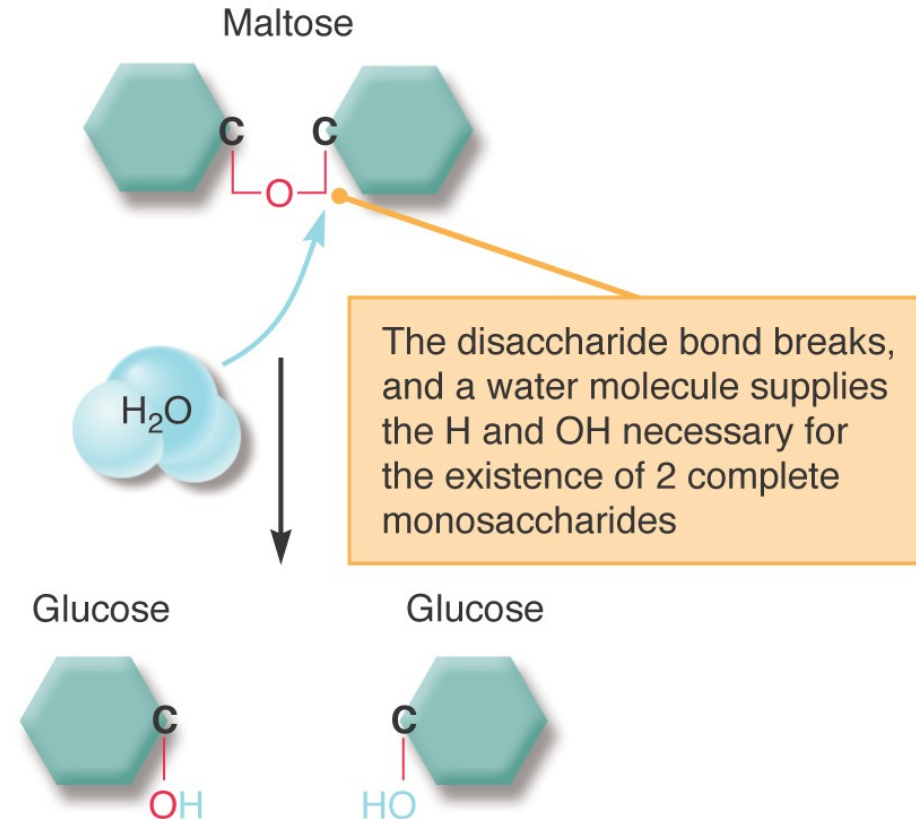


Joining and Cleaving Sugar Molecules

CONDENSATION



HYDROLYSIS



2. Disaccharides

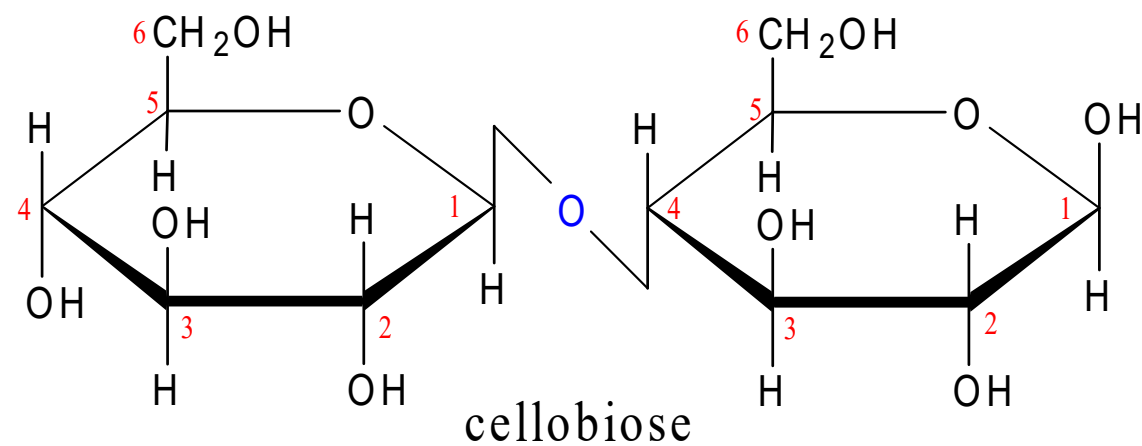
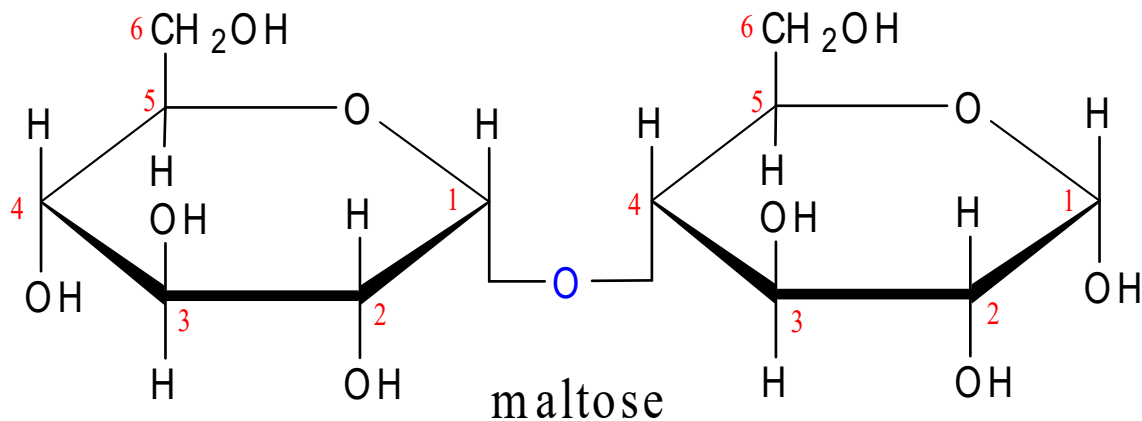
- They provide energy to human body.
- They consist of two monosaccharide units held together by glycosidic bond. So, they are glycosides.
- Most commons are *maltose*, *lactose* and *sucrose*

a) Maltose (two glucose units)

- The glycosidic linkage of maltose is symbolized as $\alpha (1 \rightarrow 4)$.
 - i.e. The anomeric carbon is bonded to oxygen on C4 of second sugar. And α -indicates the configuration of anomeric carbon atoms of both glucose units.
- Systematic name for maltose is O- α -D glucopyranosyl-(1 $\rightarrow$ 4)- α -D glucopyranose.
- Maltose is a *reducing sugar* because anomeric carbon of second glucose is free.

Disaccharides:

Maltose, a cleavage product of starch (e.g., amylose), is a disaccharide with an $\alpha(1 \rightarrow 4)$ glycosidic link between C1 - C4 OH of 2 glucoses. It is the α anomer (C1 O points down).



Cellobiose, a product of cellulose breakdown, is the otherwise equivalent β anomer (O on C1 points up).

The $\beta(1 \rightarrow 4)$ glycosidic linkage is represented as a zig-zag, but one glucose is actually **flipped over** relative to the other.

Source for maltose

- Maltose is present in germinating cereals and in barley.
- Commercial malt sugar contains maltose
- may be formed during the hydrolysis of starch.

b)Lactose

Structure

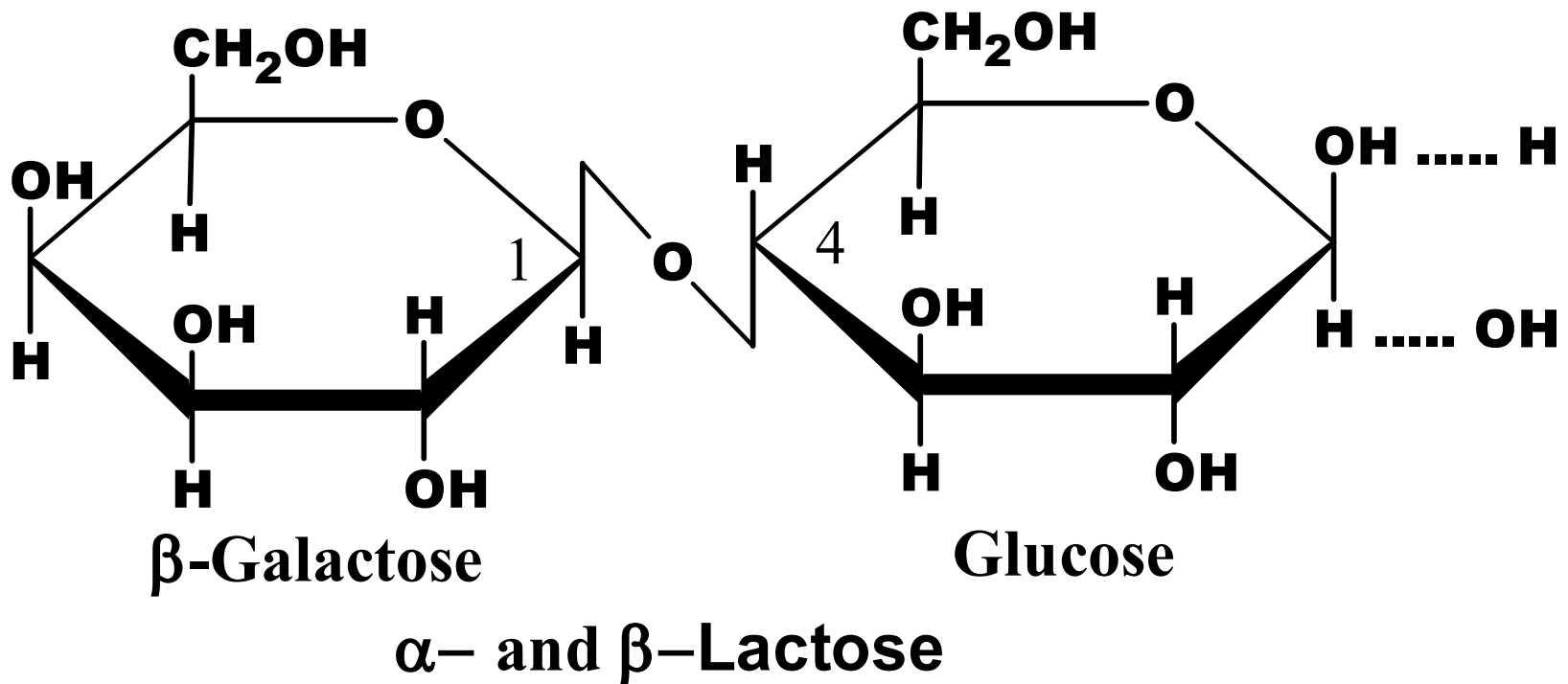
- It contains one **glucose** and one **galactose**.
- It is symbolized as β (1 $\rightarrow$ 4).
- *O- β -D galactopyranosyl-(1 $\rightarrow$ 4)- β -D-glucopyranose*.
- It is a reducing sugar because anomeric carbon of glucose is free.

Source for lactose

- Lactose is synthesized in mammary gland and hence it occurs in milk.

Lactose

- Galactose + glucose linked 1-4'.
- “Milk sugar.”



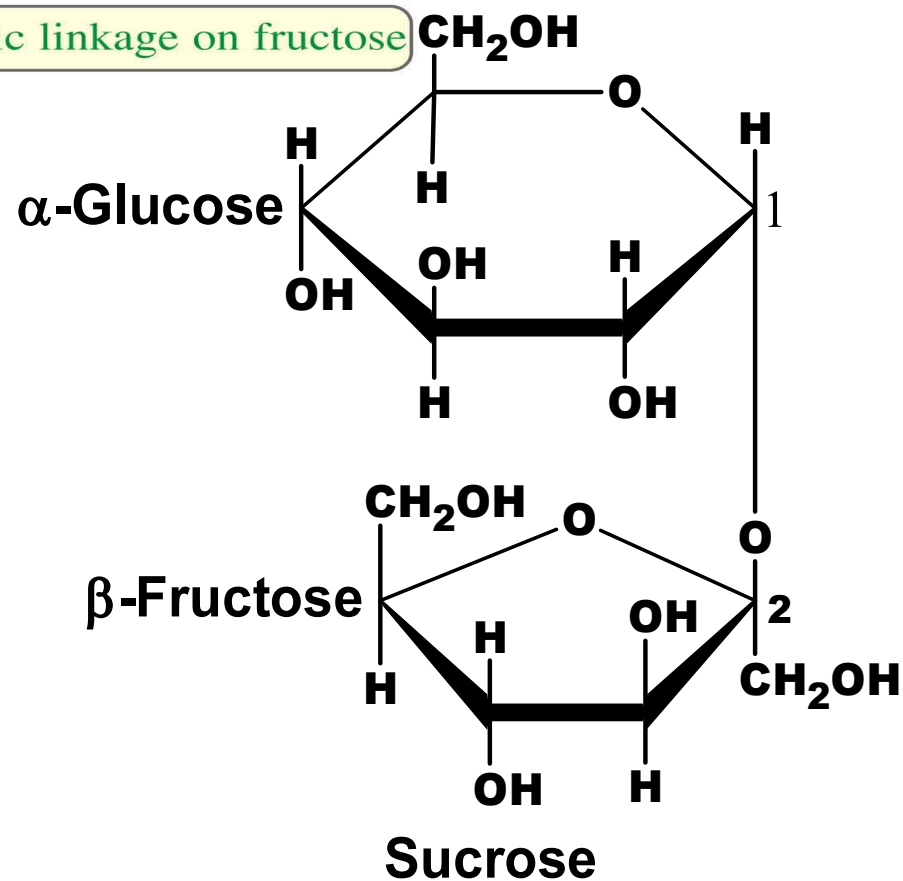
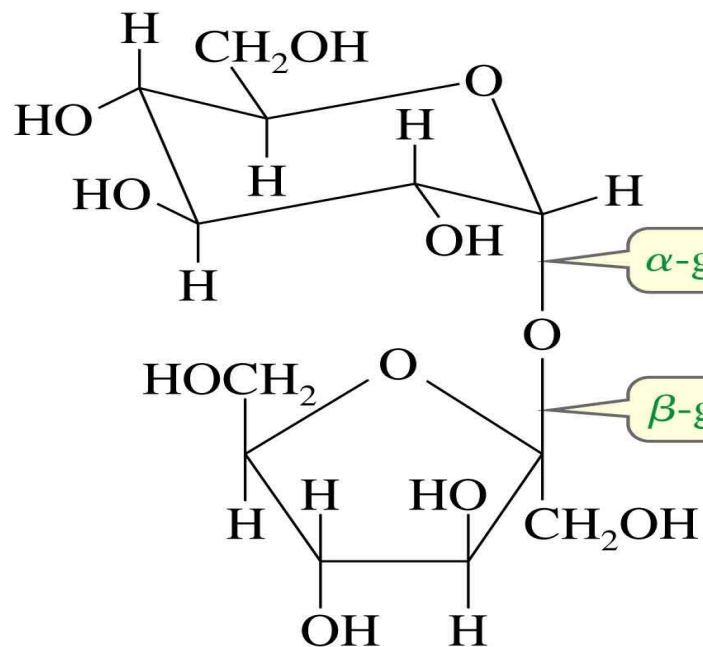
c) Sucrose

- It contains glucose and fructose.
- common table sugar.
- glucose is in α -form whereas fructose is in β -form in sucrose.
- $\alpha, \beta(1 \rightarrow 2)$.
- *O- α -D-glucopyranosyl-(1 $\rightarrow$ 2)- β -D-fructofuranose.*
- **Nonreducing sugar** b/c both the functional groups of glucose and fructose are involved in glycosidic linkage.

Source of sucrose

- Ripe fruit juices like pineapple, sugar cane, juice and honey are rich sources for sucrose.
- It also occurs in juices of sugar beets, carrot roots and sorghum.

*Sucrose, α -D-glucopyranosyl- β -D-fructofuranoside
(or β -D-fructofuranosyl- α -D-glucopyranoside)*



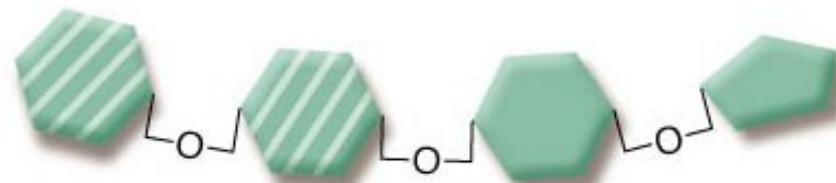
Oligosaccharides

- Beans and peas contain some oligosaccharides.
- These oligosaccharides contain 4 to 5 monosaccharide units.
- Stachyose and verbascose are a few such oligosaccharides.
- Usually these oligosaccharides are not utilized in human body.
- Also found in glycoproteins where they have important functions. Oligosaccharides are also important constituents of glycolipids present in cell membrane

Complex Carbohydrates

Oligosaccharides

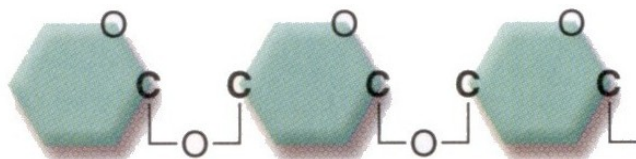
- short carbohydrate chains of 3 - 10 monosaccharides
 - found in legumes and human milk
 - Examples:
 - raffinose
 - stachyose
- } cannot be broken down by human enzymes, though can be digested by colonic bacteria



Stachyose

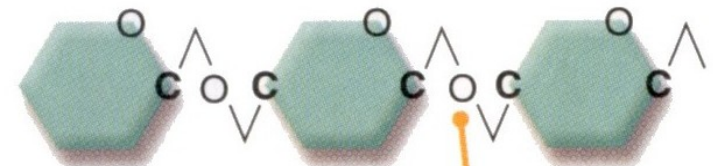
Polysaccharides

- They are polymers of monosaccharides.
- They contain more than ten monosaccharide units.
- The monosaccharides are joined together by glycosidic linkage.
 - alpha (α) bonds (starch)
 - beta (β) bonds (found in fiber)



Starch

Alpha bonds between glucose molecules in starch are easily broken by human digestive enzymes



Cellulose

Beta bonds in dietary fibers are impervious to human enzymes

Classification of Polysaccharides

- **Two types on the** basis of the type of monosaccharide present.

a) Homopolysaccharides

- They are entirely made up of one type of monosaccharides.
- On hydrolysis, they yield only one kind of monosaccharide.

b) Heteropolysaccharides

- They are made up of more than one type of monosaccharides.
- On hydrolysis they yield more than one type of monosaccharides.

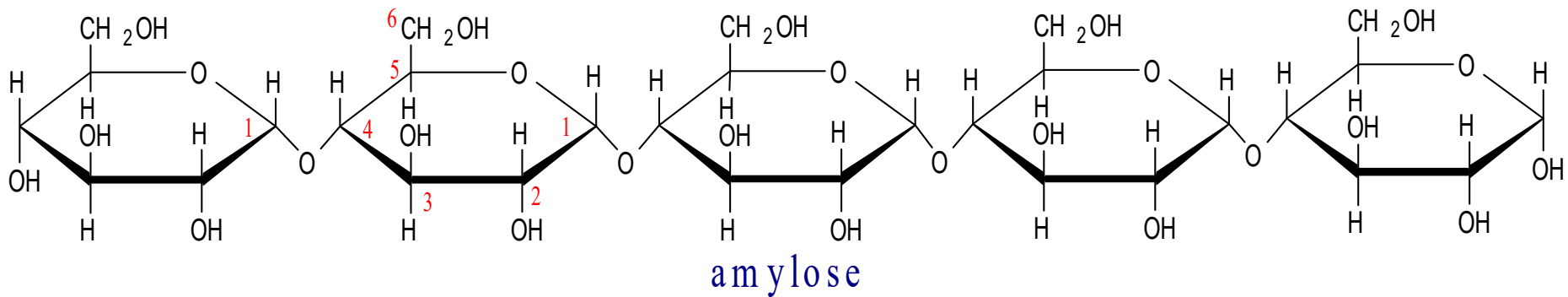
Homopolysaccharides

- Important homopoly-saccharides are starch, glycogen, cellulose.
- All these contain glucose as repeating unit.
- Other name for homopolysaccharides are homoglycans.

Starch

Structure

- It consist of two parts.
- A minor amylose component and a major amylopectin component.



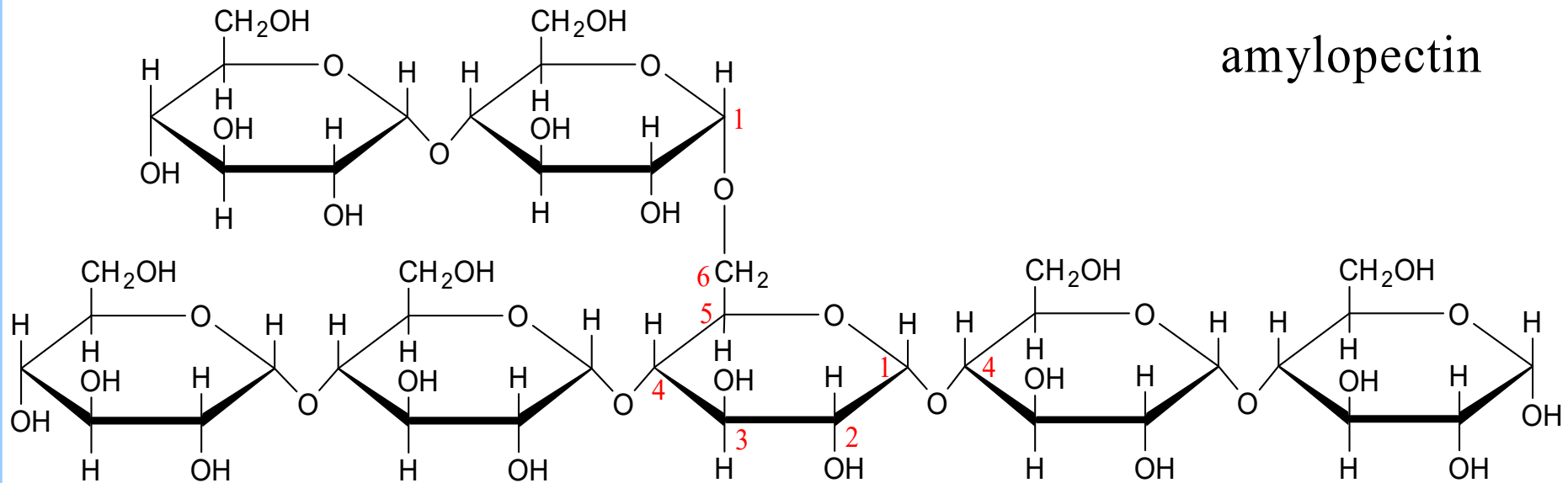
Polysaccharides:

Plants store glucose as **amylose** or **amylopectin**, glucose polymers collectively called starch.

Glucose storage in **polymeric** form **minimizes osmotic effects**.

Amylose is a glucose polymer with **$\alpha(1\rightarrow4)$** linkages.

The end of the polysaccharide with an anomeric C1 not involved in a glycosidic bond is called the **reducing end**.



Amylopectin is a glucose polymer with mainly $\alpha(1 \rightarrow 4)$ linkages, but it also has **branches** formed by $\alpha(1 \rightarrow 6)$ linkages. Branches are generally longer than shown above.

The branches produce a compact structure & provide multiple chain ends at which enzymatic cleavage can occur.

Function

1. It is the major polysaccharide present in our food.
2. It is also called as storage polysaccharide because it serve as reserve food material in plants.
3. It is present in food grains, tubers and roots like rice, wheat, potato and vegetables.

Complex Carbohydrates

Glycogen

- highly branched chains of glucose units
- animal storage form of carbohydrate
 - found in **LIVER** and **MUSCLE**
 - Humans store ~ 100g in liver; ~ 400g in muscle
- negligible source of carbohydrate in the diet (meat)



Glycogen

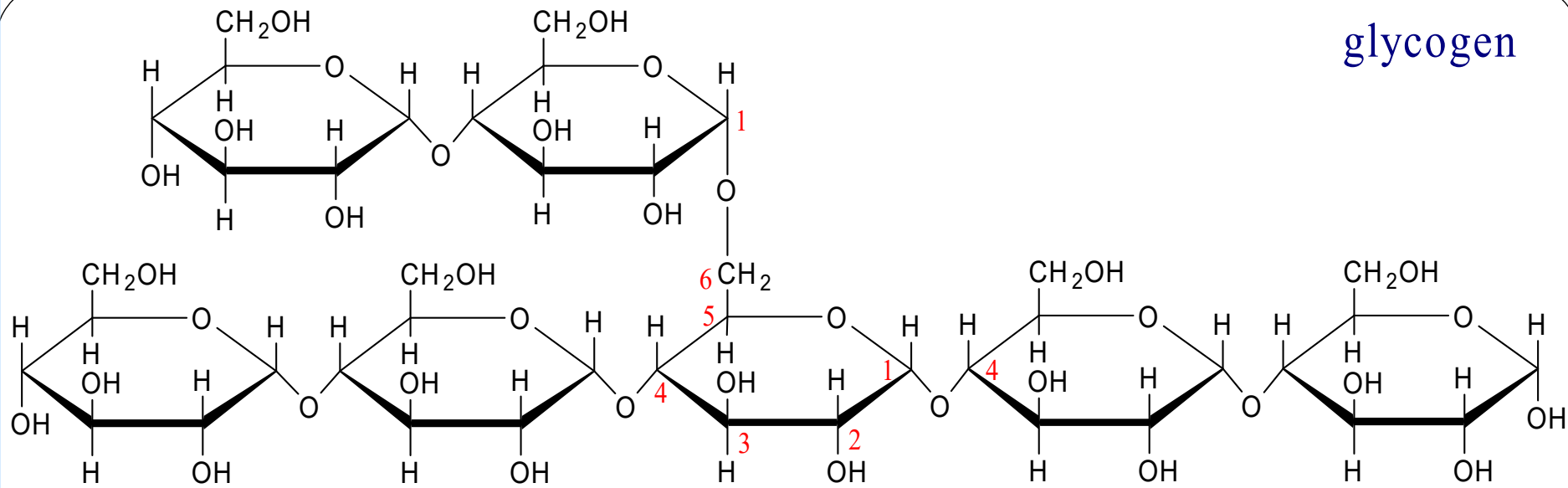
Glycogen

- similar to that of amylopectin of starch. But the number of branches in glycogen molecule is much more than amylopectin.
- There is one branch point for 6-7 glucose residues.

Function

1. It is the major storage polysaccharide (carbohydrate) in human body.
2. It is mainly present in liver and muscle.
3. It is also called as animal starch.

glycogen



Glycogen, the glucose storage polymer in **animals**, is similar in structure to amylopectin.

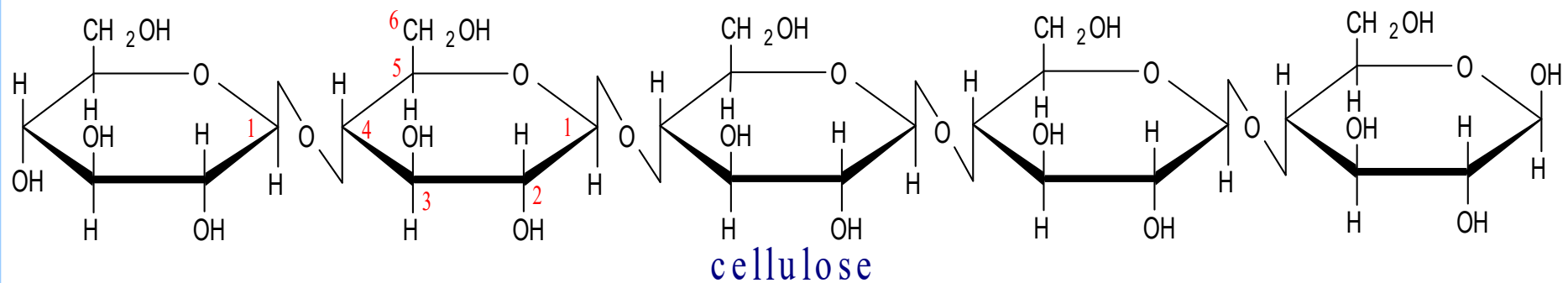
But glycogen has **more $\alpha(1\rightarrow6)$ branches**.

The highly branched structure permits rapid glucose release from glycogen stores, e.g., in muscle during exercise.

The ability to rapidly mobilize glucose is more essential to animals than to plants.

Cellulose

- It has linear chain of glucose residues, which are linked by $\beta(1 \rightarrow 4)$ glycosidic linkage.
- It occurs as bundle of fibres in nature.
- The linear chains are arranged side by side and hydrogen bonding between adjacent stands stabilizes the structure.
- The **role** of cellulose is to impart strength and rigidity to plant cell walls, which can withstand high hydrostatic pressure gradients. Osmotic swelling is prevented.



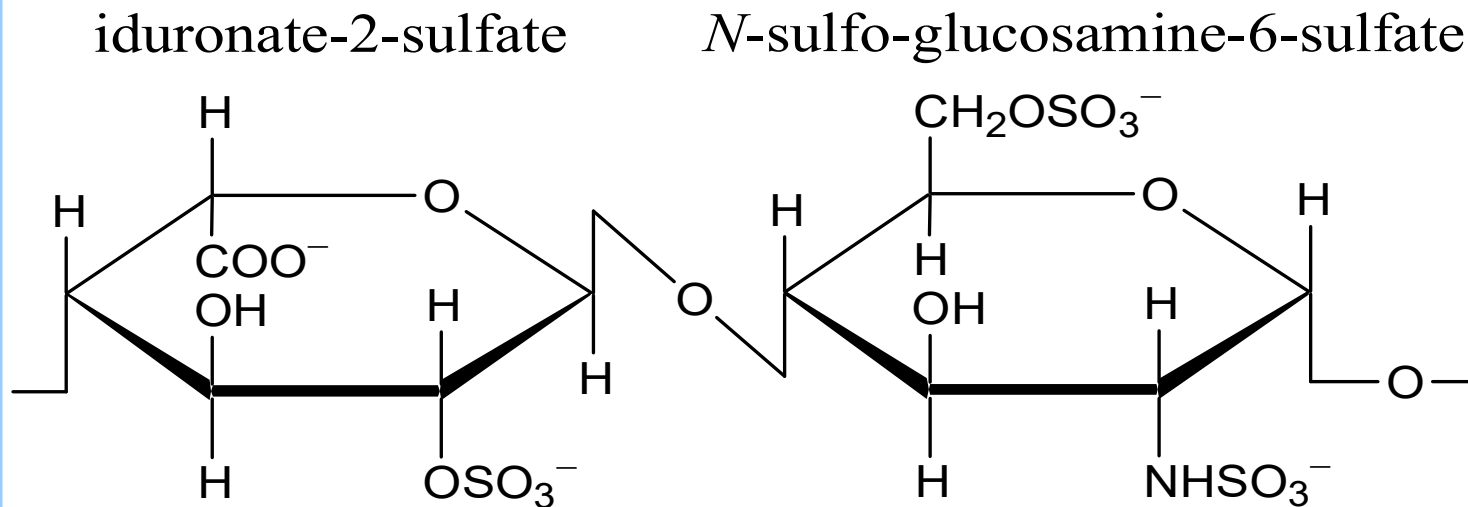
Starch	Glycogen	Cellulose
1.Nature: Stored form of carbohydrate in plants.	Stored form of carbohydrates in animals.	Structural form of carbohydrate in plant cells but prevents constipation in human.
1.Source: Cereals, e.g., wheat, rice, and tubers, e.g., potatoes.	Muscles and liver	Linen and cotton are nearly pure cellulose.
1.Solubility: Amylose is water soluble and amylopectin is insoluble.	Water soluble forming colloidal solution.	Water insoluble.
1.Nature of the chains: Amylose is helical straight chain (α -glucose units linked by α -1,4-glucosidic bonds). Amylopectin is branched chain (α -glucose units linked by α -1,4- and α -1,6-glucosidic bonds).	Branched chain similar to amylopectin but its trees are shorter and have more branches than amylopectin tree.	Straight chain (large number of β -glucose units linked by β -1,4- glucosidic bonds).
1.Reaction with iodine: Amylose gives blue color and amylopectin gives red color.	Gives red color.	No color.
1.Digestibility: Is hydrolyzed by HCl or amylase into dextrins and maltose.	Digestible by amylase into dextrins and maltose.	Non-digestiblebut HCl hydrolysis gives cellobiose.

HETEROPOLYSACCHARIDES

- also called *mucopolysaccharides* & *glycosaminoglycans*.
- Mucopolysaccharides consist of repeating disaccharide units (of two types monosaccharides.)
- mucopolysaccharides are component of connective tissue, hence often called structural polysaccharides.
- Mucopolysaccharides are also components of extracellular matrix of bone, cartilage and tendons.
- The complex of mucopolysaccharide and protein is called as *proteoglycan*.
- *Mucopolysaccharides* also function as lubricants and shock absorbers

Heparin

- The repeating disaccharide unit of heparin consist of glucosamine and either iduronic acid or glucuronic acid.
- Majority of uronic acids are iduronic acids. Further amino groups of glucosamine is sulfated.



Functions

1. Heparin is a normal anti-coagulant present blood.
2. It is produced by mast cells present in the arteries, liver, lung and skin.
3. Unlike other glycosaminoglycans, heparin is an intracellular component.
4. It can be used during the treatment of **myocardial infarction** as well as for the prevention of **deep venous thrombosis** during hospitalizations.

Complex Carbohydrates

Fiber

Dietary Fiber

- non-digestible carbohydrates (chains of monosaccharides) and lignin that are intact and intrinsic in plants (includes oligosaccharides)

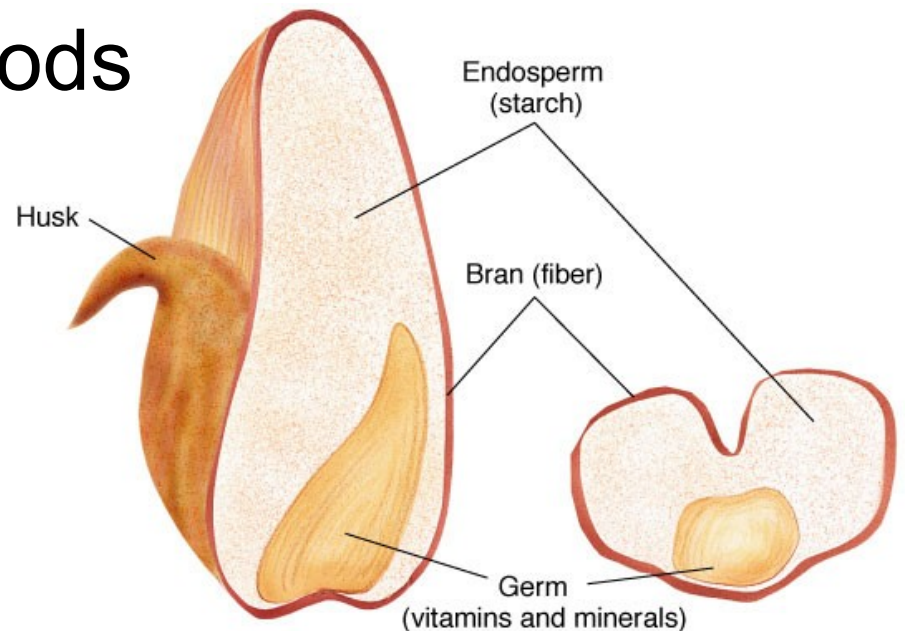
Functional Fiber

- isolated, non-digestible carbohydrates that have beneficial physiological effects in humans

Complex Carbohydrates

Fiber cont.

- dietary fiber found in all types of plant foods
- **refining** removes fiber from whole grains and other foods



Complex Carbohydrates

Fiber cont.

- types of non-starch polysaccharides include:

cellulose

hemicelluloses

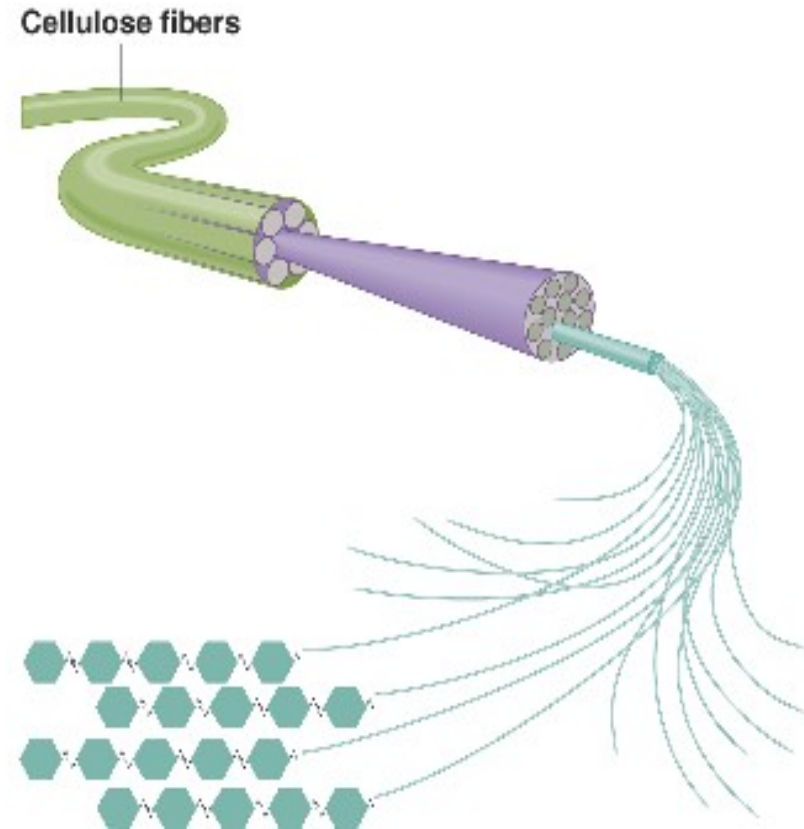
pectins

gums & mucilages

β -glucans

chitin & chitosan

lignans



Summery

Functions of Carbohydrates

1) Energy

- glucose fuels the work of most of the body's cells
 - preferred fuel of NERVOUS TISSUE (the brain, nerves) and RED BLOOD CELLS (RBC)
- excess glucose is stored as GLYCOGEN in liver and muscle tissue

Functions of Carbohydrates

2) Sparing Body Protein

- if diet does not provide enough glucose, then other sources of glucose must be found
- if carbohydrate intake **< 50 - 100 g**, body protein will be used to make glucose
- an **adequate supply of carbohydrate spares body proteins from being broken down** to synthesize glucose

Functions of Carbohydrates

3) Preventing Ketosis (Anti-ketogenic)

- carbohydrates required for the complete metabolism of fat
- incomplete fat metabolism produces **KETONES**
- an adequate supply of carbohydrate (> 50 – 100 g per day) prevents KETOSIS

Fiber

- beneficial for weight control by contributing to satiety & delay gastric emptying
- **soluble fibers** lower blood cholesterol to help reduce risk of cardiovascular disease
- minimizes risk of and helps control Type II Diabetes
- **insoluble fibers** help promote intestinal health by enlarging stool size and easing passage of stool

Soluble Fiber

- **examples** include gums, pectins, mucilages, some hemicelluloses
- **functions:**
 - delay gastric emptying
 - slow transit through the digestive system
 - delay glucose absorption
 - bind to bile, help decrease cholesterol
- **food sources:** fruits

Insoluble Fiber

- **examples** include cellulose, hemicellulose
- **functions:**
 - speed transit through the digestive tract
 - delay glucose absorption
 - increase fecal weight and soften stool to ease passage
 - reduces risk of hemorrhoids, diverticulitis and appendicitis
- **food sources:** cereal grains, legumes, vegetables, nuts

Fiber: Too much of a good thing?

Excessive amounts of fiber may lead to:

- displacement of other foods in the diet
- intestinal discomfort
- interference with the absorption of other nutrients

Summary

- **Carbohydrates** are major constituents of human food and human tissues. They are characterized by the type and number of monosaccharide residues in their molecules.
- **Glucose** is the major metabolic fuel of humans and a universal fuel of the fetus.
- The physiologically important monosaccharides include **glucose**, the "blood sugar," and **ribose**, an important constituent of nucleotides and nucleic acids.
- The important disaccharides include **maltose** (glucosyl-glucose), **sucrose**(glucosyl-fructose), and **lactose** (galactosyl-glucose), .
- **Starch** and **glycogen** are storage polymers of glucose in plants and animals, respectively. Starch is the major metabolic fuel in the diet.

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Thank You!!